

POOJA HEGDE

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SUMMARY

Full-stack software engineer and MCA graduate specializing in AI-integrated web applications. Built and shipped four end-to-end projects spanning a CNN-based medical diagnostic tool (~90% test accuracy), a secure full-stack SaaS-style QR code generator, and algorithm visualization and inventory management systems — using React, Node.js, Express.js, MySQL, Python, and TensorFlow. Passionate about building scalable, user-focused software and continuously improving problem-solving skills.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, SQL

Web & Frameworks: React, Node.js, Express.js, Flask, HTML, CSS, Bootstrap, Tailwind CSS, Vite

ML & Data: TensorFlow, Keras, OpenCV, MySQL

Tools & Other: Git, GitHub, Linux, VS Code, JWT Authentication, Bcrypt, Xampp, Postman

PROJECTS

NephroScan AI – AI-Powered Chronic Kidney Disease Detection

Technologies: Python, TensorFlow, Keras, OpenCV, Flask, React

- Built an end-to-end AI-powered web application for Chronic Kidney Disease detection, **training a CNN on CT scan images** and achieving **~90% classification accuracy** on the test set.
- Implemented image preprocessing and **Grad-CAM visualizations** to deliver explainable, confidence-scored predictions rather than opaque black-box outputs.
- Built a responsive React interface for secure image upload and **real-time diagnosis**, demonstrating a full data-to-deployment AI workflow.

QR Forge – Full-Stack QR Code Generator

Technologies: React, TypeScript, Vite, Tailwind CSS, MySQL, JWT Authentication, Bcrypt

- Built a secure full-stack QR code generator supporting **8 formats**: Text, URL, Email, Phone, Wi-Fi, Contact (vCard), WhatsApp, and Google Maps.
- Implemented **JWT-based authentication** and **REST APIs** with Express.js, encrypted user credentials with **Bcrypt**, and persisted QR generation history in MySQL.
- Designed a responsive React interface with reusable components, **QR customization**, logo embedding, **PNG/SVG export**, clipboard support, and dark/light theming; QR codes generate in under **1 second** locally.

AlgoViz – Interactive Algorithm Visualizer

Technologies: HTML, CSS, JavaScript, Bootstrap

- Built an interactive web app that animates **8 classic algorithms** in real time — Bubble Sort, Linear Search, Bucket Sort, Binary Search, Merge Sort, Selection Sort, Fractional Knapsack, and Sum of Subsets — to make step-by-step algorithm behavior visible and easier to understand.

Rental Equipment Management System

Technologies: PHP, HTML, CSS, JavaScript, Bootstrap, MySQL

- Built a full-stack web application to manage **equipment inventory**, customer bookings, and **admin operations** through a secure dashboard.
- Implemented user **authentication**, booking management, and an admin dashboard for real-time inventory tracking.

EDUCATION

Dayananda Sagar College of Arts, Science and Commerce, Bangalore, Karnataka

Feb 2024 – Nov 2025

Master of Computer Applications (MCA), CGPA: 7.61/10

JMJ BCA College, Sirsi, Karnataka (Karnataka University Dharwad)

Sep 2020 – Oct 2023

Bachelor of Computer Applications (BCA), CGPA: 8.61/10

CERTIFICATIONS

- AWS Cloud Practitioner Essentials (Sep 2025)
- Programming in Python — Meta (Jul 2025)
- Java Foundations — Oracle Academy (May 2024)